Webhook Ingestion & Retry Mechanism Hub

Functional and Non functional requirements:

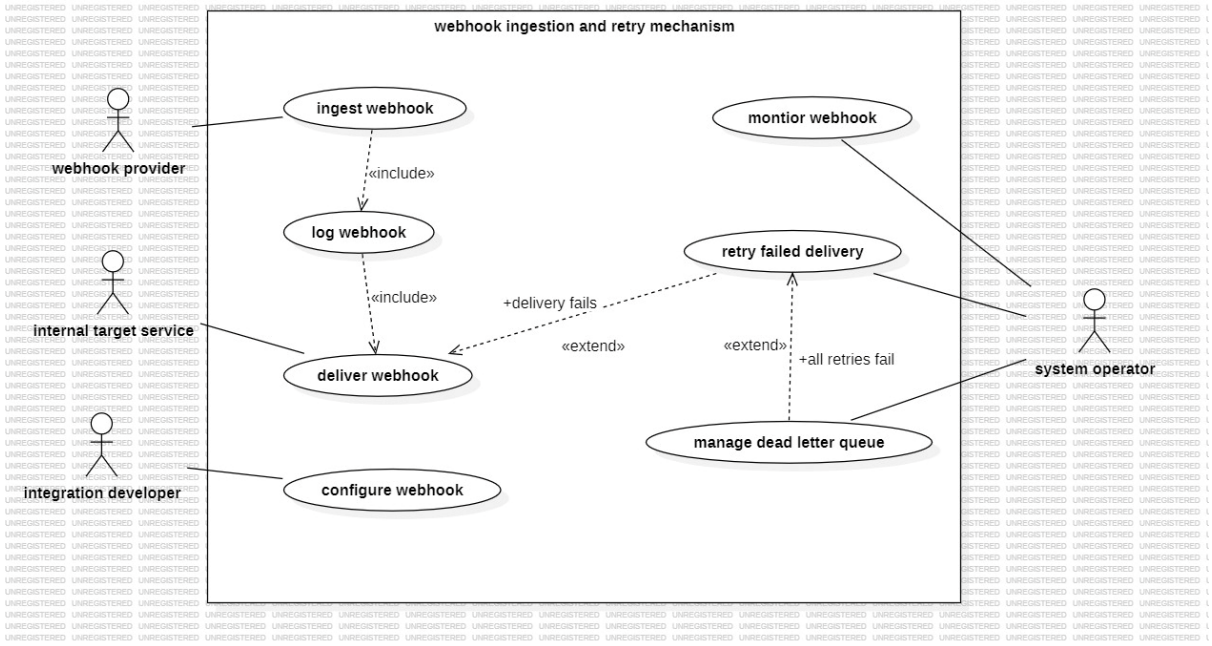
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-0 01	Functional	The system shall ingest inbound webhook requests from registered third-party webhook providers.	High	**Pass:** A valid webhook sent by a registered provider is received by the system and acknowledged for processing. **Fail:** The webhook is rejected, lost, or not accepted for processing.	Webhook ingestion is the fundamental function of the system	
FR-0 02	Functional	The system shall log the payload and relevant delivery information for every successfully ingested webhook.	High	**Pass:** After a webhook is successfully ingested, a corresponding log entry containing the webhook payload and delivery information is available. **Fail:** No corresponding log entry is created, or the recorded information is incomplete.	Logging is important for tracking webhook events and troubleshooting delivery failures.	
FR-0 03	Functional	The system shall forward each ingested webhook payload to	High	**Pass:** When a valid webhook is ingested and the	Core purpose of the gateway	

		its configured target endpoint.		target endpoint is available, the target endpoint receives the same webhook payload. **Fail:** The target endpoint does not receive the webhook or receives an incorrect payload.		
FR-0 04	Functional	The system shall automatically retry failed webhook deliveries using exponential backoff.	High	**Pass:** When a webhook delivery fails, the system automatically attempts redelivery using the configured exponential-backoff intervals of **1s, 2s, 4s, and 8s**. **Fail:** The system does not retry a failed delivery, retries without the configured backoff intervals, or permanently drops the webhook without further handling.	Automatic retries improve delivery reliability when the target service is temporarily unavailable. The exponential backoff also prevents the system from repeatedly overwhelming a failing target.	
FR-0 05	Functional	The system shall persist webhook events that remain undelivered after the configured retry	High	**Pass:** After all configured retry attempts fail, the webhook is stored in the	The DLQ ensures that permanently failed webhook	

		attempts in a dead-letter queue.		dead-letter queue and can be identified as an undelivered event. **Fail:** The webhook is discarded, lost, or not persisted after all retry attempts fail.	events are preserved for later investigation or recovery.	
NFR001	Nonfunc.	The ingestion endpoint shall buffer inbound webhooks with **99.999% reliability** and support throughput of up to **2,500 requests/second**.	High	**Pass:** Under simulated peak load, the system processes up to **2,500 webhook requests/second** while maintaining at least **99.999% successful ingestion reliability**. **Fail:** The system cannot sustain 2,500 requests/second or ingestion reliability falls below 99.999%.	The webhook hub must remain reliable and capable of handling a high volume of incoming third-party events.	
NFR002	Nonfunc.	The system shall authenticate incoming webhook requests before accepting them for processing.	High	**Pass:** A webhook request with valid authentication credentials is accepted, while a request with invalid or missing credentials is rejected.	The webhook endpoint is exposed to third-party systems, so the system needs to prevent unauthorized parties from submitting	

				Fail: An unauthenticated or incorrectly authenticated webhook is accepted for processing	arbitrary webhook events.	

Use case diagram:



Use case flow:

Goal	**Ingest and Deliver Webhook**
Actor	**Webhook Provider with support from the Internal Target Service**
Main Flow	1. The Webhook Provider sends a webhook event to the system. 2. The system receives and ingests the webhook. 3. The system logs the webhook payload and delivery information. 4. The system identifies the configured target endpoint for the webhook. 5. The system forwards the webhook payload to the Internal Target Service. 6. The Internal Target Service successfully receives and processes the webhook. 7. The system records the successful delivery. 8. Webhook processing is completed successfully.
Alternate Flow	Failed Webhook Delivery At Step 5, if the target endpoint fails to receive the webhook: 5a. The system detects the delivery failure. 5b. The system automatically retries the delivery using exponential backoff: 1s → 2s → 4s → 8s. 5c. If a retry succeeds, the system records the successful delivery and the flow continues to Step 7. 5d. If all configured retry attempts fail, the system stores the webhook in the Dead-Letter Queue (DLQ). 5e. The failed webhook is preserved for the System Operator to inspect/manage.